BAB 2. TCP/IP Model

2.1 Tujuan

- Mahasiswa mampu melakukan identifikasi transmisi data menggunakan model TCP/IP
- Mahasiswa mampu melakukan identifikasi layer dari model TCP/IP
- Mahasiswa mampu menggunakan aplikasi untuk mengidentifikasi data dari model TCP/IP

2.2 Dasar Teori

2.2.1 OSI Layer

OSI adalah referensi komunikasi dari Open System Interconnection. OSI model digunakan sebagai referensi untuk membahas spesifikasi protokol. OSI model terdiri dari 7 layer seperti pada tabel 1.

Layer #	Nama	Unit Enkapsulasi	Deskripsi/Fungsi
7	Application	Data	Mendefinisikan pelayanan (services) komunikasi pada suatu aplikasi
6	Presentation	Data	Mendefinisikan format data
5	Session	Data	Mendefinisikan memulai, pengontrolan dan mengakhiri komunikasi
4	Transport	Segment	Mendefinisikan jenis pengiriman
3	Network	Paket	Mendefinisikan pengiriman dari titik ke titik.
2	Data Link	Frame	Mendefinisikan cara pengaturan pengiriman data
1	Physical	bit	Mendifinisikan bentuk interface dari sebuah media transmisi

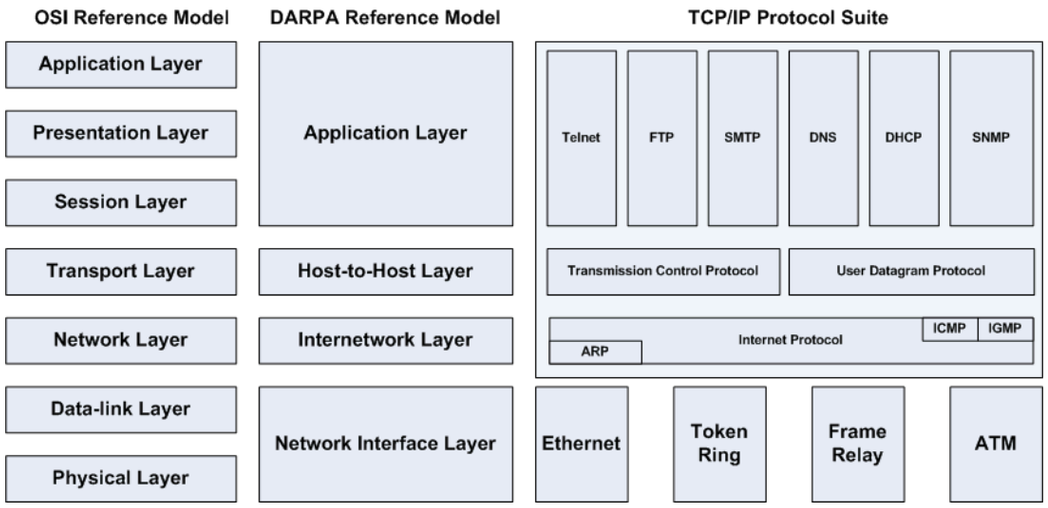
Tabel 1. OSI Layer

Dimana setiap layer memiliki fungsi dan contoh masing-masing.

2.2.2 TCP/IP Model

TCP/IP (singkatan dari Transmission Control Protocol/Internet Protocol), adalah gabungan dari protokol TCP dan IP sebagai sekelompok protokol yang mengatur komunikasi data dalam proses tukar-menukar data dari satu komputer ke komputer lain di dalam jaringan Internet yang akan memastikan pengiriman data sampai ke alamat yang dituju. Protokol ini tidaklah dapat berdiri sendiri, karena memang protokol ini berupa kumpulan protokol (Protocol Suite).

Arsitektur TCP/IP tidaklah berbasis OSI Layer, tetapi menggunakan model referensi DARPA. Seperti pada gambar 11., TCP/IP mengimplementasikan arsitektur berlapis yang terdiri atas empat lapis. Empat lapis ini, dapat dipetakan terhadap model OSI layer. Empat lapis ini, disebut juga DARPA Model atau DOD Model.



Gambar 11. OSI Layer vs TCP/IP Protocol Suite.

2.3 Pengenalan Aplikasi

2.3.1 Wireshark

Wireshark adalah aplikasi penganalisa paket jaringan (Network Packet Analyzer). Sebuah software penganalisa paket jaringan yang bekerja dengan cara menangkap paket jaringan dan mencoba untuk menampilkan data paket yang tertangkap dengan detail.
Wireshark bersifat open source. Contoh
penggunaan wireshark:
- Administrator jaringan menggunakan untuk melakukan penangan masalah jaringan
- Network security engineer menggunakan untuk memeriksa keamanan jaringan
- Developer menggunakan untuk melakukan debug pada implementasi protokol
- Umum digunakan untuk mempelajari internal protokol jaringan

Wireshark memiliki fitur:
- Tersedia untuk UNIX dan Windows.
- Menangkap paket data langsung dari network interface.
- Tampilan paket dengan informasi protokol yang sangat rinci.
- Open dan Save data paket yang dicapture.
- Import dan Eksport paket data dari dank e banyak program capture lainnya.
- Menyaring paket data dengan berbagai macam kriteria.
- Warnai layar paket berdasarkan filter.

2.3.2 Cara penggunaan Wireshark

Langkah-langkah menggunakan wireshark

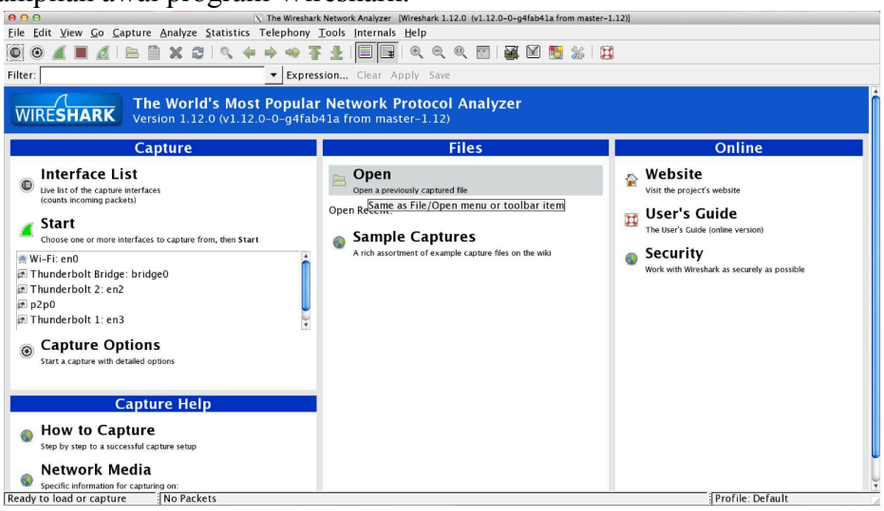
1. Klik ikon…



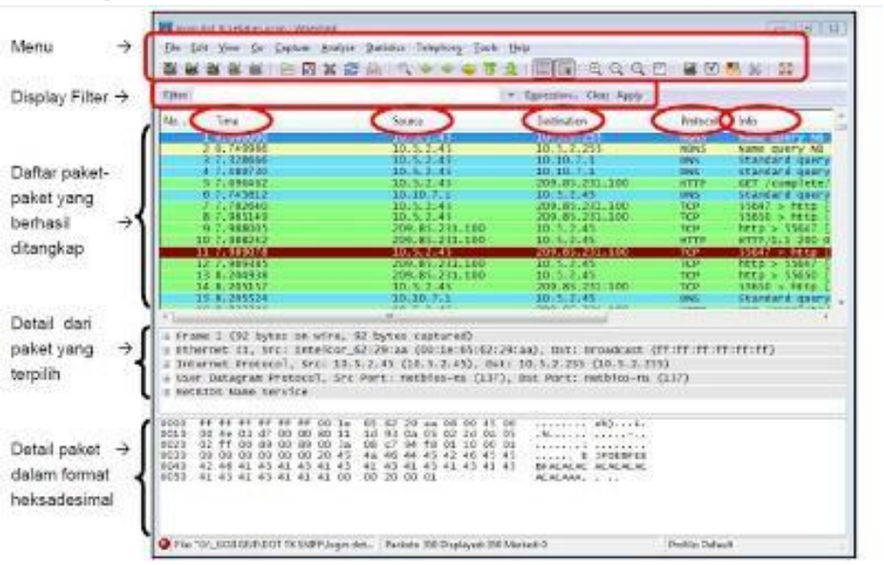
2. Kemudian akan tampil aplikasi wireshark dengan diawali spash screen.



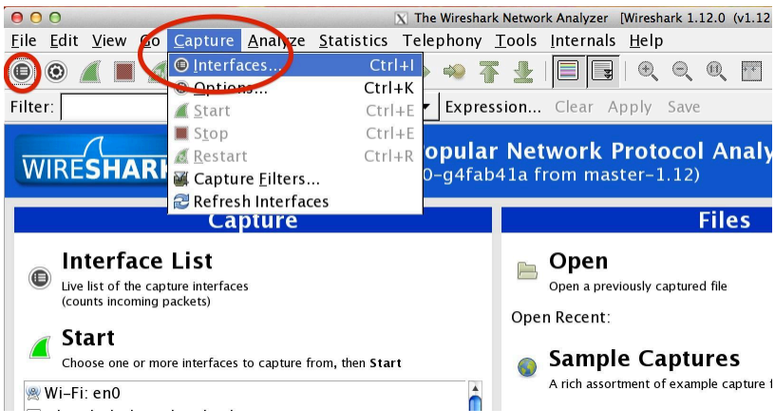
3. Tampilan awal program Wireshark.



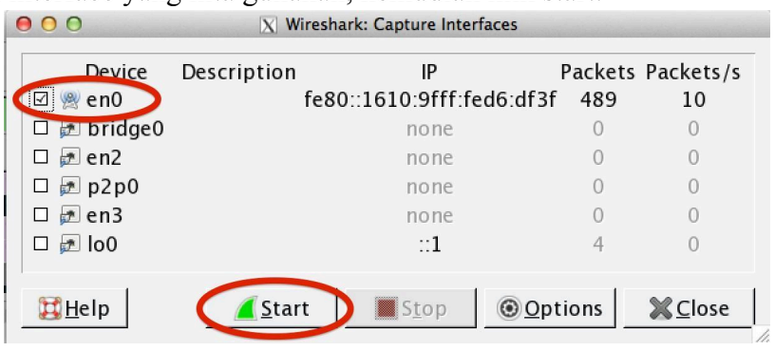
4. Keterangan menu wireshark.



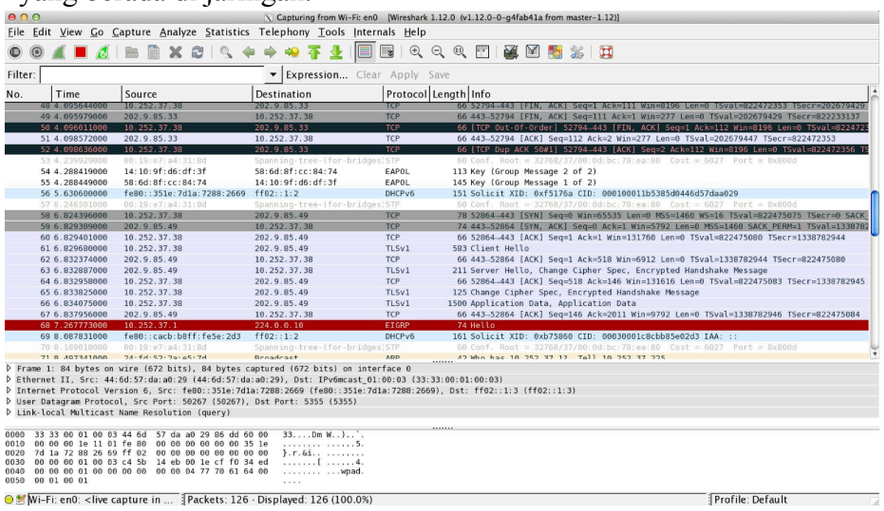
5. Melakukan CAPTURE data menggunakan wireshark. Pilih menu INTERFACE.



6. Pilih interface yang kita gunakan, kemudian klik start.



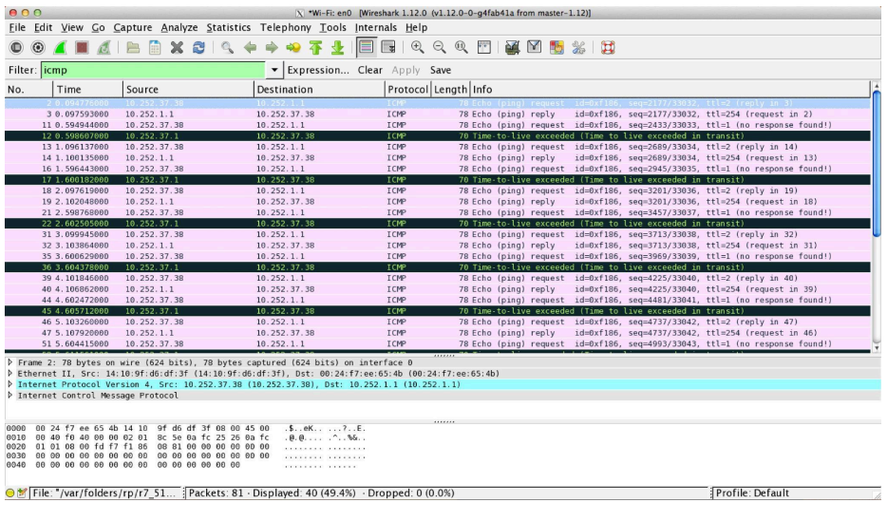
7. Setelah mengaktifkan interfaces, Wireshark akan beraksi membaca semua data yang berada di jaringan.



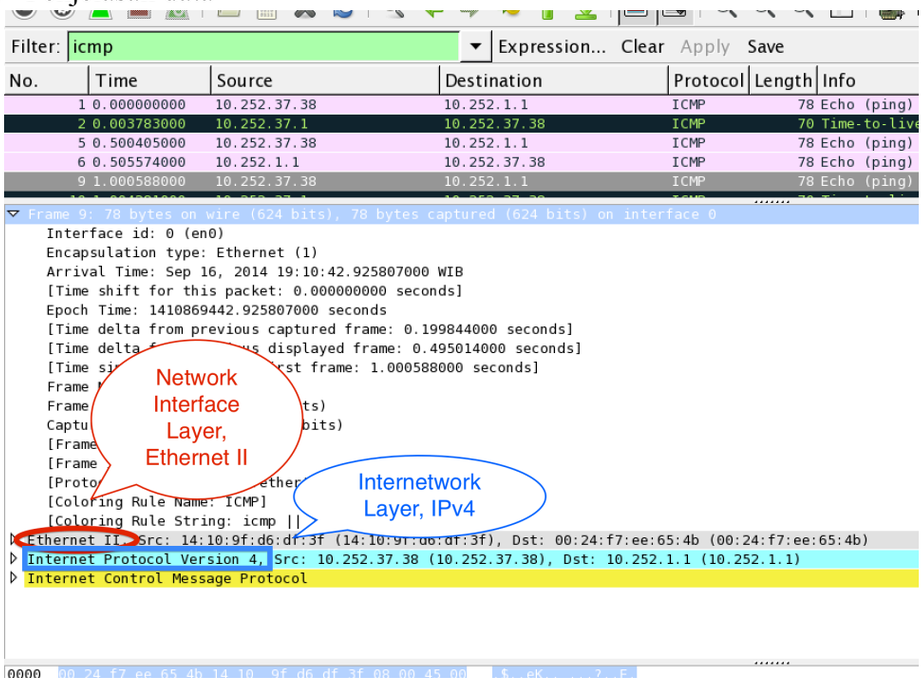
Terlihat banyak data yang dapat dicapture.
Keterangan:

Source	Alamat IP pengirim
Destination	Alamat IP tujuan
Protocol	Jenis protokol yang digunakan, seperti TCP, DHCP, dll
Length	Panjang data yang dikirimkan
Info	Informasi isi data yang dikirimkan

8. Ketikkan ICMP pada filter box untuk melakukan penyaringan data yang akan ditampilkan sehingga hanya data PING yang akan ditampilkan.



9. Penjelasan data



2.4 Persiapan Praktikum

1. Modul praktikum
2. PC yang dilengkapi dengan aplikasi Wireshark
3. Jaringan aktif
4. Lembar Laporan Praktikum (Lampiran)
5. Koneksi LAN dan Internet

2.5 Langkah-langkah Praktikum

2.5.1 Menjalankan aplikasi wireshark

1. Mahasiswa melakukan pengecekan apakah pada PC yang dipakai sudah terinstall aplikasi wireshark ?!!

2. Apabila belum terinstall, mahasiswa harus melakukan installasi di PC yang digunakan !!!

   Untuk OS Linux Debian/Ubuntu: dhoto@mypc $
   su
   root@mypc # apt-get install wireshark

   Untuk OSX:
   - Download wireshark dan XQuartz
   - Lakukan konfigurasi XQuartz pada wireshark

   Untuk Windows:
   - Download wireshark
   - Klik 2x pada installer wireshark

3. Klik ikon Wireshark !!!
4. Klik pada menu CAPTURE dan memilih INTERFACES.
5. Pilih interface yang akan digunakan untuk mengrekam data.
6. Klik Start.
7. Pada OS linux, jalankan aplikasi TERMINAL dan ketik perintah PING yang digunakan untuk melakukan pengecekan jaringan (tanpa tanda $):
   $ ping 10.252.1.1
8. Kembali ke aplikasi wireshark, hentikan rekam data dengan cara klik tombol merah pada wireshark. Ambil data untuk jenis tipe: icmp dengan cara memfilter di aplikasi wireshark dan lakukan screen shot pada tampilan wireshark yang kalian gunakan.

2.5.2 Melakukan rekam data untuk jenis protokol tertentu

1. Lakukan praktikum seperti kegiatan 2.5.1 dari nomer 4 s/d 8.
2. Ganti kegiatan pada no 7, dengan kegiatan berikut:
   a. Jalankan aplikasi terminal dan ketik perintah NSLOOKUP yang digunakan untuk mengetahui IP dari suatu tujuan (tanpa tanda $):
      $ nslookup www.pens.ac.id
      Ambil data untuk jenis tipe: dns
   b. Jalankan aplikasi web browser dan arahkan ke alamat: ftp://newfs.pens.ac.id
      Download salah satu data yang ada di alamat tersebut !!!
      Ambil data untuk jenis tipe: ftp
   c. Jalankan aplikasi web browser dan arahkan ke alamat: http://www.pens.ac.id
      Ambil data untuk jenis tipe: http
3. Untuk masing-masing kegiatan diatas, laporkan data yang telah direkam sesuai dengan layer pada TCP/IP model. Gunakan lembar **Laporan praktikum 2. TCP/IP Model**. (Lampiran)

2.6 Lampiran

Laporan praktikum 2. TCP/IP Model

Kegiatan : Analisis Data DNS
Nama : Novia Syabillah Kahlizah
NRP : 5325600049

Data layer

Application layer

```
Domain Name System (query)
    Transaction ID: 0xc9e4
    Flags: 0x0100 Standard query
        0... .... .... .... = Response: Message is a query
        .000 0... .... .... = Opcode: Standard query (0)
        .... ..0. .... .... = Truncated: Message is not truncated
        .... ...1 .... .... = Recursion desired: Do query recursively
        .... .... .0.. .... = Z: reserved (0)
        .... .... ...0 .... = Non-authenticated data: Unacceptable
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    Queries
        www.notion.so: type A, class IN
            Name: www.notion.so
            [Name Length: 13]
            [Label Count: 3]
            Type: A (1) (Host Address)
            Class: IN (0x0001)
    [Response In: 258]
```

Host-to-host layer

```
User Datagram Protocol, Src Port: 56832, Dst Port: 53
    Source Port: 56832
    Destination Port: 53
    Length: 39
    Checksum: 0x0102 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 133]
    [Stream Packet Number: 1]
    [Timestamps]
        [Time since first frame: 0.000000000 seconds]
        [Time since previous frame: 0.000000000 seconds]
    UDP payload (31 bytes)
```

Internetwork layer

```
Internet Protocol Version 4, Src: 10.252.129.214, Dst: 10.252.128.2
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 59
    Identification: 0x05be (1470)
    000. .... = Flags: 0x0
        0... .... = Reserved bit: Not set
        .0.. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x1d24 [validation disabled]
```

```
    [Header checksum status: Unverified]
    Source Address: 10.252.129.214
    Destination Address: 10.252.128.2
    [Stream index: 82]
```

Network Interface layer

```
Frame 253: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface
\Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546}, id 0
    Section number: 1
    Interface id: 0 (\Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546})
        Interface name: \Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546}
        Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: Oct 13, 2026 15:39:43.024841800 SE Asia Standard Time
    UTC Arrival Time: Oct 13, 2026 08:39:43.024841800 UTC
    Epoch Arrival Time: 1791880783.024841800
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 28.249600 milliseconds]
    [Time since reference or first frame: 1.668799100 seconds]
    Frame Number: 253
    Frame Length: 73 bytes (584 bits)
    Capture Length: 73 bytes (584 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:udp:dns]
    Character encoding: ASCII (0)
    [Coloring Rule Name: UDP]
    [Coloring Rule String: udp]

Ethernet II, Src: LiteonTechno_a7:88:b1 (24:b2:b9:a7:88:b1), Dst: VMware_9f:5a:c2 (00:50:56:9f:5a:c2)
    Destination: VMware_9f:5a:c2 (00:50:56:9f:5a:c2)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: LiteonTechno_a7:88:b1 (24:b2:b9:a7:88:b1)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 95]
```

2.7 Lampiran

Laporan praktikum 2. TCP/IP Model

Kegiatan : Analisis Data HTTP
Nama : Novia Syabillah Kahlizah
NRP : 5325600049

Data layer

Application layer

```
Hypertext Transfer Protocol
    HTTP/1.1 301 Moved Permanently\r\n
        Response Version: HTTP/1.1
        Status Code: 301
        [Status Code Description: Moved Permanently]
        Response Phrase: Moved Permanently
    Server: nginx\r\n
    Date: Tue, 13 Oct 2026 08:54:04 GMT\r\n
    Content-Type: text/html\r\n
    Content-Length: 162\r\n
        [Content length: 162]
    Connection: keep-alive\r\n
    Location: https://www.pens.ac.id/\r\n
    \r\n
    [Request in frame: 962]
    [Time since request: 17.704200 milliseconds]
    [Request URI: /]
    [Full request URI: http://www.pens.ac.id/]
    File Data: 162 bytes

Line-based text data: text/html (7 lines)
    <html>\r\n
    <head><title>301 Moved Permanently</title></head>\r\n
    <body>\r\n
    <center><h1>301 Moved Permanently</h1></center>\r\n
    <hr><center>nginx</center>\r\n
    </body>\r\n
    </html>\r\n
```

Host-to-host layer

```
Transmission Control Protocol, Src Port: 80, Dst Port: 62360, Seq: 1, Ack: 79, Len: 353
    Source Port: 80
    Destination Port: 62360
    [Stream index: 8]
    [Stream Packet Number: 6]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 353]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 4286508785
    [Next Sequence Number: 354    (relative sequence number)]
    Acknowledgment Number: 79    (relative ack number)
    Acknowledgment number (raw): 3365450347
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
```

```
    .... ...1 .... = Acknowledgment: Set
    .... .... 1... = Push: Set
    .... .... .0.. = Reset: Not set
    .... .... ..0. = Syn: Not set
    .... .... ...0 = Fin: Not set
    [TCP Flags: ·····AP···]
  Window: 502
  [Calculated window size: 64256]
  [Window size scaling factor: 128]
  Checksum: 0xefa5 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  [Timestamps]
    [Time since first frame in this TCP stream: 59.941800 milliseconds]
    [Time since previous frame in this TCP stream: 2.800 microseconds]
  [SEQ/ACK analysis]
    [iRTT: 25.907800 milliseconds]
    [Bytes in flight: 353]
    [Bytes sent since last PSH flag: 353]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (353 bytes)
```

Internetwork layer

```
Internet Protocol Version 4, Src: 202.9.85.176, Dst: 10.252.129.214
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 393
  Identification: 0xb482 (46210)
  010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 63
  Protocol: TCP (6)
  Header Checksum: 0xd960 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 202.9.85.176
  Destination Address: 10.252.129.214
  [Stream index: 131]
```

Network Interface layer

```
Frame 964: Packet, 407 bytes on wire (3256 bits), 407 bytes captured (3256 bits) on interface
\Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546}, id 0
  Section number: 1
  Interface id: 0 (\Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546})
    Interface name: \Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546}
    Interface description: Wi-Fi
  Encapsulation type: Ethernet (1)
  Arrival Time: Oct 13, 2026 15:54:04.759440500 SE Asia Standard Time
  UTC Arrival Time: Oct 13, 2026 08:54:04.759440500 UTC
  Epoch Arrival Time: 1791881644.759440500
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 2.800 microseconds]
  [Time delta from previous displayed frame: 17.704200 milliseconds]
  [Time since reference or first frame: 4.760612600 seconds]
  Frame Number: 964
  Frame Length: 407 bytes (3256 bits)
  Capture Length: 407 bytes (3256 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
```

```
    [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
Ethernet II, Src: VMware_b5:55:03 (00:50:56:b5:55:03), Dst: LiteonTechno_a7:88:b1 (24:b2:b9:a7:88:b1)
    Destination: LiteonTechno_a7:88:b1 (24:b2:b9:a7:88:b1)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: VMware_b5:55:03 (00:50:56:b5:55:03)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 195]
```

2.8 Lampiran

Laporan praktikum 2. TCP/IP Model

Kegiatan : Analisis Data FTP
Nama : Novia Syabillah Kahlizah
NRP : 5325600049

Data layer

Application layer

```
File Transfer Protocol (FTP)
    RETR Paramore - All I Wanted.mp3\r\n
        Request command: RETR
        Request arg: Paramore - All I Wanted.mp3
[Current working directory: /audio/Paramore]
[Command response frames: 889]
[Command response bytes: 3605845]
[Command response first frame: 70606]
[Command response last frame: 71925]
[Response duration: 527ms]
[Response bitrate: 54737Kbps]
[Setup frame: 70550]
```

Host-to-host layer

```
Transmission Control Protocol, Src Port: 60734, Dst Port: 21, Seq: 136, Ack: 462, Len: 34
    Source Port: 60734
    Destination Port: 21
    [Stream index: 156]
    [Stream Packet Number: 24]
    [Conversation completeness: Incomplete, DATA (15)]
        ..0. .... = RST: Absent
        ...0 .... = FIN: Absent
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ··DASS]
    [TCP Segment Len: 34]
    Sequence Number: 136    (relative sequence number)
    Sequence Number (raw): 421189995
    [Next Sequence Number: 170    (relative sequence number)]
    Acknowledgment Number: 462    (relative ack number)
    Acknowledgment number (raw): 1796060340
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····AP···]
    Window: 254
    [Calculated window size: 65024]
    [Window size scaling factor: 256]
    Checksum: 0xa9d8 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
        [Time since first frame in this TCP stream: 114.828300 milliseconds]
        [Time since previous frame in this TCP stream: 131.000 microseconds]
    [SEQ/ACK analysis]
```

```
    [This is an ACK to the segment in frame: 70551]
    [The RTT to ACK the segment was: 131.000 microseconds]
    [iRTT: 17.298200 milliseconds]
    [Bytes in flight: 34]
    [Bytes sent since last PSH flag: 34]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (34 bytes)
```

Internetwork layer

```
Internet Protocol Version 4, Src: 10.252.129.214, Dst: 202.9.85.80
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 74
  Identification: 0xda70 (55920)
  010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x7411 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.252.129.214
  Destination Address: 202.9.85.80
  [Stream index: 27]
```

Network Interface layer

```
Frame 70552: Packet, 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface
\Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546}, id 0
  Section number: 1
  Interface id: 0 (\Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546})
    Interface name: \Device\NPF_{0F3776D6-9696-46C9-AE9C-EDD7B2F15546}
    Interface description: Wi-Fi
  Encapsulation type: Ethernet (1)
  Arrival Time: Oct 13, 2026 16:37:54.084702900 SE Asia Standard Time
  UTC Arrival Time: Oct 13, 2026 09:37:54.084702900 UTC
  Epoch Arrival Time: 1791884274.084702900
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 131.000 microseconds]
  [Time delta from previous displayed frame: 131.000 microseconds]
  [Time since reference or first frame: 10 minutes, 15.906513600 seconds]
  Frame Number: 70552
  Frame Length: 88 bytes (704 bits)
  Capture Length: 88 bytes (704 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:ftp]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP]
  [Coloring Rule String: tcp]
Ethernet II, Src: LiteonTechno_a7:88:b1 (24:b2:b9:a7:88:b1), Dst: VMware_b5:55:03
(00:50:56:b5:55:03)
  Destination: VMware_b5:55:03 (00:50:56:b5:55:03)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
```

```
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
Source: LiteonTechno_a7:88:b1 (24:b2:b9:a7:88:b1)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 43]
```